ONE WEEK SHORT TERM COURSE ON

"Large Language Models: A Mathematical and Practical Approach"

Indian Institute of Information Technology Kottayam

Deadline for Registration: May 15, 2026

18th to 22nd May 2026 (Hybrid Mode)

About: This course is designed to provide a comprehensive and balanced understanding of both the theoretical foundations and practical applications of modern Natural Language Processing (NLP). It introduces essential mathematical concepts, including probability, N-gram models, entropy, and perplexity, which underpin language modeling. The course progresses to deep learning techniques such as neural networks, word embeddings, and sequence models including RNNs and LSTMs, along with attention mechanisms and transformer architectures that power contemporary large language models. Emphasis is placed on practical skills, including optimization, fine-tuning through transfer learning and LoRA, prompt engineering, and retrieval-augmented generation (RAG). The course also examines real-world applications, key evaluation metrics such as BLEU and ROUGE, and the risks and limitations associated with large language models. It concludes with a mini project and demonstration, enabling participants to apply their knowledge and gain hands-on experience in building and deploying LLM-based solutions.

Target Audience:

Students and Researchers: Undergraduate and postgraduate students in computer science, data science, engineering, and related disciplines.

Educators and Academicians: Faculty members who wish to strengthen their understanding of modern language model architectures and applications.

Industry Professionals: Data scientists, AI/ML engineers, and technology professionals looking to build, fine-tune, and deploy LLM-based solutions.

Technology Enthusiasts and Practitioners: Individuals with a basic background in programming and mathematics who are interested in gaining practical and theoretical knowledge of large language models and their applications.

Course content:

- Mathematical Foundations of LLM
- Deep Learning for NLP
- Transformer Models & Architectures
- Model Training & Fine-Tuning
- Prompt Engineering & RAG
- Evaluation & Applications
- Limitations & Ethics
- Hands-on Implementation

Sessions:

All sessions will be conducted in hybrid mode. Each day will consist of three interactive sessions (2 lectures 90 mins each and one hands-on activities of 120 minutes).

Organized by

Department of Computer Science and Engineering,

Indian Institute of Information Technology Kottayam **(IIIT Kottayam)**

In association with

Computational Engineering and Data Modelling Research Group and FACTS-H Lab

All attendees will be provided with lecture slides, source code, and e-certificates.

Course Coordinator
Dr. Jayakrushna Sahoo,
Email: jsahoo@iiitkottayam.ac.in,
Phone: +91 (0) 482-2202164 , +91-7872456553

Registration Fees:

Offline Registration fee: Rs 2000/-
Online Registration fee: Rs: 1500/-

Registration Form:
https://forms.gle/3whK3p9ZwKVGokFVA

Steps to make Payment:

1. Visit SBI Collect: **https://onlinesbi.sbi.bank.in/sbicollect/**
2. Select Educational Institute
3. Search for IIIT Kottayam
4. Select Payment category -STC LLM
5. Enter the necessary details and make the payment